Status	Finished
Started	Sunday, 11 October 2026, 1:02 AM
Completed	Sunday, 11 October 2026, 1:06 AM
Duration	3 mins 50 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

- a. O👎
- b. O(1) ✓
- c. O(n log n)
- d. O(log n)

Question **2**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

- a. Jump search
- b. Hashing
- c. Binary search
- d. Linear search ✓

Question **3**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

- a. In the left sublist
- b. In the middle of the list
- c. At the beginning of the list
- d. In the right sublist ✓

Question **4**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

- a. Binary search ✓
- b. Linear search
- c. Hash search
- d. Sequential search

Question **5**

Correct

Mark 1.00 out of 1.00

What is the first step in binary search?

- a. Sort the list
- b. Compare the target element with the first element in the list
- c. Compare the target element with the middle element in the list ✓
- d. Divide the list into two equal parts

Question **6**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

- a. The search continues until the end of the list
- b. The search stops immediately ✓
- c. The search starts over from the beginning
- d. The search backtracks to find duplicate elements

Question **7**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

- a. Linear search requires the list to be sorted.
- b. Linear search divides the list into halves.
- c. Linear search can be applied to both sorted and unsorted lists. ✓
- d. Linear search is more efficient than binary search on large lists.

Question **8**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

- a. Binary search algorithm is not efficient when the data elements more than 1500 ✓
- b. There must be a mechanism to access middle element directly
- c. Must use a sorted array
- d. Requirement of sorted array is expensive when a lot of insertion and deletions are needed

Question **9**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

- a. When the item is somewhere in the middle of the array ✓
- b. When the item is not the array at all
- c. Item is the last element in the array or item is not there at all
- d. When the item is the last element in the array

Question **10**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

- a. Binary search
- b. Dynamic search
- c. Hashing
- d. Linear search ✓

Question **11**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Binary search

b. Both (1) & (3)

c. Hash search

d. Linear search ✓

Question **12**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 1000

b. 500

c. 10 ✓

d. 100

Question **13**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search has a lower time complexity on large, sorted lists ✓

b. Binary search can find multiple instances of the target element

c. Binary search does not require dividing the list

d. Binary search works on unsorted lists

Question **14**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. n/2
b. log n
c. 1
d. n ✓

Question **15**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The search ends successfully ✓
b. The search continues in the right sublist
c. The list is sorted
d. The search continues in the left sublist